Yakeen Plus (2025)

Practice Test - 15

DURATION : 200 Minutes | **DATE : 16/12/2024** | **M. MARKS : 720**

Topics Covered

Physics :	Gravitation, Properties of Solids and Liquids, Thermodynamics, Kinetic Theory of Gases, Oscillation and Waves
Chemistry :	Redox Reaction, Electrochemistry, Hydrocarbon, Haloalkanes and Haloarenes, Coordination Compound
Biology :	**(Botany) :** Morphology of Flowering Plants, Anatomy of Flowering Plants, Respiration in Plants, Photosynthesis in Higher Plants, Plant Growth and Development **(Zoology) :** Body Fluids and Circulation, Excretory Products & their Elimination, Locomotion & Movement, Neural Control & Coordination Chemical Coordination & Integration

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min**. duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Botany)** & Section IV **(Zoology)** and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : ______________________________

Roll Number : ______________________________

OMR Bar Code Number : ______________________________

Candidate's Signature : ____________________ **Invigilator's Signature** ____________________

SECTION-(I) PHYSICS

SECTION - A

1. A spaceship is launched into a circular orbit close to earth's surface. What additional velocity has now to be imparted to the spaceship in the orbit to overcome the gravitational pull? (Take, radius of earth = 6400 km, $g = 9.8$ m/s^2)

(1) 3.28 km/s (2) 12 km/s
(3) 10 km/s (4) 40 km/s

2. What is the maximum height attained by a body projected with a velocity equal to one-third of the escape velocity from the surface of the earth?
(Take, radius of the earth = R)

(1) $\frac{R}{2}$ (2) $\frac{R}{3}$
(3) $\frac{R}{5}$ (4) $\frac{R}{8}$

3. Two satellites S_1 and S_2 are revolving round a planet in coplanar circular orbits of radii r_1 and r_2 in the same direction, respectively. Their respective periods of revolution are 1 h and 8 h. The radius of orbit of satellite S_1 is equal to 10^4km. What will be their relative speed (in km/h) when they are closest?

(1) $\frac{\pi}{2}\times 10^4$ (2) $\pi \times 10^4$
(3) $2\pi \times 10^4$ (4) $4\pi \times 10^4$

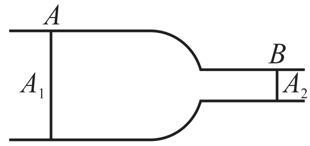
4. The length of a metal wire is l_1, when the tension in it is T_1 and length is l_2 when the tension is T_2. The natural length of the wire is

(1) $\frac{l_1 + l_2}{2}$ (2) $\sqrt{l_1 l_2}$
(3) $\frac{l_1T_2 - l_2T_1}{T_2 - T_1}$ (4) $\frac{l_1T_2 + l_2T_1}{T_2 + T_1}$

5. A force F is applied on the wire of radius r and length L and change in the length of the wire is l. If the same force F is applied on the wire of the same material and radius $2r$ and length $2L$, then the change in length of the other wire is:

(1) l (2) $2l$
(3) $\frac{l}{2}$ (4) $4l$

6. Water is flowing in a non-viscous tube as shown in the diagram. The area of cross-section at point A and B are A_1 and A_2, respectively. The pressure difference between point A and B is Δp. The rate of flow is:



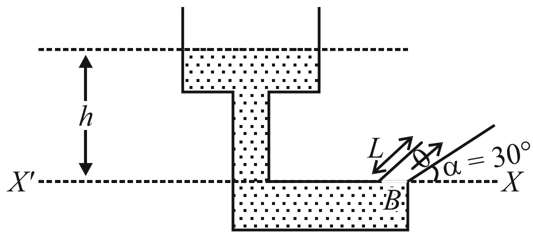
(1) $A_1A_2\sqrt{\frac{2\Delta p}{\rho\left(A_1^2 - A_2^2\right)}}$

(2) $A_1A_2\sqrt{\frac{\Delta p}{\rho\left(A_1^2 - A_2^2\right)}}$

(3) $A_1A_2\sqrt{\frac{2\Delta p}{A_1^2 - A_2^2}}$

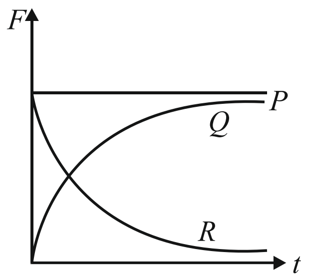
(4) None of the above

7. Determine the height above the dashed line XX' attained by the water stream coming out through the hole situated at point B in the diagram given below. Given that $h = 10$ m, $L = 2$ m and $\alpha = 30°$.



(1) 10 m (2) 7.1 m
(3) 5 m (4) 3.2 m

8. A spherical ball is dropped in a long column of viscous liquid. Which of the following graphs represent the variation of
(i) gravitational force with time
(ii) viscous force with time
(iii) net force acting on the ball with time?

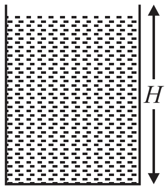


(1) Q, R, P (2) R, Q, P
(3) P, Q, R (4) R, P, Q

9. A hole is made at the bottom of the tank filled with water (density 1000 kg/m^3). If the total pressure at the bottom of the tank is 3 atmosphere (Take, 1 atmosphere = 10^5 N/m^2), then the velocity of efflux is:

(1) $\sqrt{200}$m/s (2) $\sqrt{400}$m/s
(3) $\sqrt{500}$m/s (4) $\sqrt{800}$m/s

10. A cylindrical vessel is filled with water as shown in figure. A hole should be bored so that the water comes out upto maximum distance from the surface.



(1) $\frac{3H}{4}$ (2) $\frac{H}{4}$

(3) H (4) $\frac{H}{2}$

11. A rectangular vessel when full of water, takes 10 min to be emptied through an orifice in its bottom. How much time will it take to be emptied when half filled with water?

(1) 9 min (2) 5 min

(3) 7 min (4) 3 min

12. The black body spectrum of an object O_1 is such that its radiant intensity, (i.e., intensity per unit wavelength interval) is maximum at a wavelength of 200 nm. Another object O_2 has the maximum radiant intensity at 600 nm. The ratio of power emitted per unit area by source O_1 to that of source O_2 is:

(1) 1 : 81 (2) 1 : 9

(3) 9 : 1 (4) 81 : 1

13. A non-conducting body floats in a liquid at 20°C with $\frac{2}{3}$ of its volume immersed in the liquid. When liquid temperature is increases to 100°C. $\frac{3}{4}$, of body's volume is immersed in the liquid. Then, the coefficient of real expansion of the liquid is (neglecting the expansion of container of the liquid)

(1) 15.6×10^{-4}°C^{-1} (2) 156×10^{-4}°C^{-1}

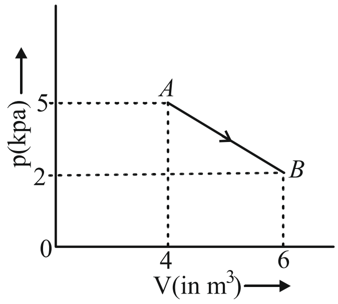
(3) 1.56×10^{-4}°C^{-1} (4) 0.156×10^{-4}°C^{-1}

14. Two slabs A and B of different materials but of the same thickness are joined end to end to form a composite slab. The thermal conductivities of A and B are K_1 and K_2 respectively. A steady tempeature difference of 12°C is maintained across the composite slab. If $K_1 = \frac{K_2}{2}$, then temperature difference across slab A is:

(1) 4°C (2) 6°C

(3) 8°C (4) 10°C

15. One mole of an ideal diatomic gas undergoes transition from A to B along a path AB as shown below. The change in internal energy of the gas during the transition is:



(1) 20 kJ (2) –12 kJ

(3) –20 kJ (4) 20 J

16. In an isobaric process of an ideal gas, the ratio of heat supplied and work done by the system i.e. $\left(\frac{Q}{W}\right)$ is:

(1) $\frac{\gamma - 1}{\gamma}$ (2) γ

(3) $\frac{\gamma}{\gamma - 1}$ (4) 1

17. A perfect gas goes from state A to state B by absorbing 8×10^5 J of heat and doing 6.5×10^5 J of external work. It is now transferred between the same two states in another process in which it absorbs 10^5 J of heat. In the second process,

(1) work done on gas is 10^5 J

(2) work done on gas is 0.5×10^5 J

(3) work done by gas is 10^5

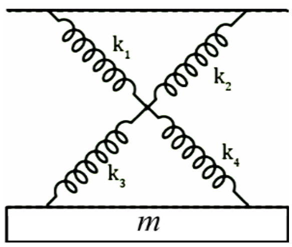
(4) word done by gas is 0.5×10^5 J

18. The latent heat of vaporisation of water is 2240 J/gm. If the work done in the process of vaporisation of 1g is 168 J, then increase in internal energy will be:

(1) 1904 J (2) 2072 J

(3) 2240 J (4) 2408 J

19. As shown in figure a simple harmonic motion oscillator having identical four springs has time period. ($k_1 = k_2 = k_3 = k_4 = k$)



(1) $T = 2\pi\sqrt{\frac{m}{4k}}$ (2) $T = 2\pi\sqrt{\frac{m}{2k}}$

(3) $T = 2\pi\sqrt{\frac{m}{k}}$ (4) $T = 2\pi\sqrt{\frac{m}{8k}}$

20. Root mean square (rms) speed of the molecules of ideal gas is v. If pressure is increased two times at constant temparature, then the rms speed will become:

(1) $\frac{v}{2}$ (2) v

(3) $2v$ (4) 4

21. A gas mixture contains one mole O_2 gas and one mole He gas, find the ratio of specific heat at constant pressure to that at constant volume of the gaseous misture.

(1) 2 (2) 1.5

(3) 2.5 (4) 4

22. If pressure of CO_2 (real gas) in a container is given by $p = \frac{RT}{2V - b} - \frac{a}{4v^2}$, then mass of the gas in container is:

(1) 11 g (2) 22 g

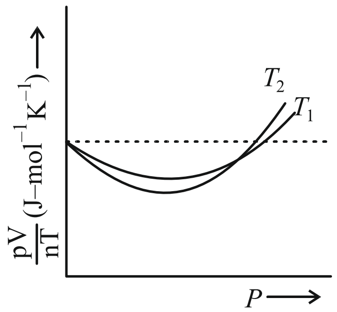
(3) 33 g (4) 44 g

23. By what percentage should the pressure of a given mass of a gas be increased, so as to decrease its volume by 10% at a constant temperature?

(1) 5% (2) 7.2%

(3) 12.5% (4) 11.1%

24. The figure below shows the plot of $\frac{pV}{nT}$ versus p for oxygen gas at two different temperatures.



Read the following statements concerning the above curves

(i) The dotted line corresponds to the 'ideal' gas behaviour.

(ii) $T_1 > T_2$

(iii) The value of $\frac{pV}{nT}$ at the point, where the curves meet on the Y-axis is the same for all gases.

Which of the above statements is **true?**

(1) (i) only

(2) (i) and (ii) only

(3) All of these

(4) None of these

25. Two paritcles of equal mass m go around a circle of radius R under the action of their mutual gravitational attraction. The speed of each particle is:

(1) $\frac{1}{2R}\sqrt{\frac{1}{Gm}}$

(2) $\sqrt{\frac{Gm}{2R}}$

(3) $\frac{1}{2}\sqrt{\frac{Gm}{R}}$

(4) $\sqrt{\frac{4Gm}{R}}$

26. If the gravitiational force has varies as $r^{-5/2}$ instead of r^{-2}, the potential energy of a particle at a distance r from the centre of the Earth would be proportional to:

(1) $r^{3/2}$

(2) $r^{-3/2}$

(3) r^{-1}

(4) r^{-2}

27. Two simple harmonic motions are represented by the equations $y_1 = 0.1\sin\left(100\pi t + \frac{\pi}{3}\right)$ and y_2 = 0.1cos πt.

The phase difference of the velocity of particle 1 with respect to the velocity of particle 2 is at $t = 0$.

(1) $\frac{-\pi}{3}$ (2) $\frac{\pi}{6}$

(3) $\frac{-\pi}{6}$ (4) $\frac{\pi}{3}$

28. The velocity vector V and displacement vector x of a particle executing SHM are related as $\frac{vdv}{dx} = -\omega^2 x$

with the initial condition $v = v_0$ at $x = 0$ the velocity v, when displacement is x, is:

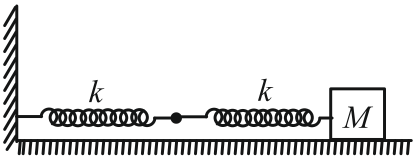
(1) $v = \sqrt{v_0^2 + \omega^2 x^2}$

(2) $v = \sqrt{v_0^2 - \omega^2 x^2}$

(3) $v = \sqrt[3]{v_0^3 + \omega^3 x^3}$

(4) $v = v_0 - (\omega^3 x^3 e^{x^3})^{1/3}$

29. Two springs are connected to a block of mass M placed on a frictionless surface as shown below. If both the springs have a springs constat k, the frequency of oscillation of block is



(1) $\frac{1}{2\pi}\sqrt{\frac{k}{M}}$ (2) $\frac{1}{2\pi}\sqrt{\frac{k}{2M}}$

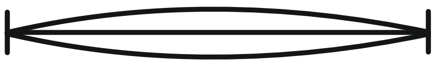
(3) $\frac{1}{2\pi}\sqrt{\frac{2k}{M}}$ (4) $\frac{1}{2\pi}\sqrt{\frac{M}{k}}$

30. Simple pendulum is executing simple harmonic motion with time period T. If the length of the pendulum is increased by 21% then the increase in the time period of the pendulum of the increased length is:

(1) 22% (2) 13%

(3) 50% (4) 10%

31. Frequency of the wave is 50 Hz. Length is 1 m and mass of string is 10 gm. What is the tension of the string?



(1) 5 N (2) 10 N

(3) 100 N (4) 50 N

32. A closed organ pipe of length 1.2 m vibrates in its first overtone mode. The pressure variation is maximum at

(1) 0.4 m from the open end

(2) 0.4 m from the closed end

(3) Both (1) and (2)

(4) 0.8 m from the open end

33. Two tuning forks P and Q when set vibrating, give 4 beats/s. If a prong of the fork P is filed, the beats are reduced to 2 Beat/s. What is frequency of P, if that of Q is 250 Hz?

(1) 246 Hz

(2) 250 Hz

(3) 254 Hz

(4) 252 Hz

34. An organ pipe closed at one end has fundamental frequency of 1500 Hz. The maximum number of overtones generted by this pipe which a normal person can hear is?

(1) 4 (2) 13

(3) 6 (4) 9

35. Velocity of sound in air is 320 m/s. A pipe closed at one end has a length of 1 m. Neglecting end corrections, the air column in the pipe can resonate for sound of frequency.

(a) 80 Hz (b) 240 Hz

(c) 320 Hz (d) 400 Hz

Choose the **correct** statements;

(1) a, b, c (2) a, b, d

(3) b, c, d (4) a, b, c, d

SECTION - B

36. The equation for the displacement of a stretched string is given by $y = 8\sin^2\pi\left[\frac{t}{0.02} - \frac{x}{100}\right]$ whree y and x are in cm and t in second

List-I		List-II	
(A)	Amplitude of wave in cm	(I)	50
(B)	Frequency of wave in Hz	(II)	4
(C)	Velocity of wave in ms^{-1}	(III)	4π
(D)	Maximum particle velocity in cm/s	(IV)	5000
		(v)	$400\ \pi$

Choose the **correct** answer from the options given below:

(1) A-II, B-I, C-I, D-V

(2) A-III, B-IV, C-I, D-V

(3) A-II, B-I, C-IV, D-V

(4) A-III, B-IV, C-IV, D-V

37. A wave disturbance in a medium is described by $y(x,t) = 0.02 \cos (50\ \pi t + \pi/2) \cos (10\pi x)$, where x and y are in metres and t in seconds.

(A) A node occurs at $x = 0.15$ m

(B) An antinode occurs at $x = 0.3$ m

(C) The speed of the wave is 5.0 m/s

(D) The wavelength is 0.2 m

(1) A, C, D (2) B, C, D

(3) A, B, C, D (4) C, D

38. Regarding speed of sound is gas match the following:

List-I		List-II	
(A)	Temperature of gas is made 4 times and pressure 2 times	(I)	Speed becomes $2\sqrt{2}$ times.
(B)	Only pressure is made 4 times without change in temparatuare	(II)	Speed becomes two times
(C)	Only temparatuare changed to 4 times	(III)	Speed remain unchanged
(D)	Molecular mass of the gas is made 4 times	(IV)	Speed become half

Choose the correct answer from the options given below:

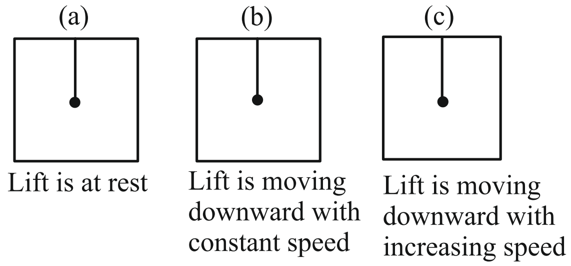
(1) A-I, B-III, C-IV, D-II

(2) A-II, B-III, C-IV, D-I

(3) A-II, B-III, C-II, D-IV

(4) A-III, B-II, C-I, D-III

39. A simple pendulum is suspended from the celling of a lift. Three cases of lift are given:

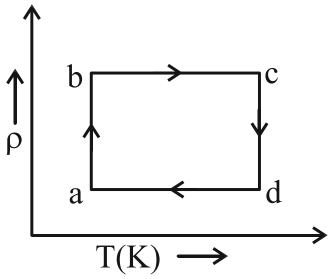


In which case time period of pendulum is:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

(1) only in (a)
(2) only in (b)
(3) only in (c)
(4) in both (a) & (b)

40. In the ρ – T graph (for fixed mass of an ideal gas), shown in the figure match the following (ρ→ density, T→ temperature)



List-I		List-II	
(A)	Process ab	(I)	isochoric process
(B)	Process bc	(II)	$\Delta U = 0$
(C)	Process cd	(III)	P increasing
(D)	Process da	(IV)	P decreasing

Choose the **correct** answer from the options given below:

(1) A-III, IV, B-I, II C-I, IV D-III, II
(2) A-II, III B-I, C-II, IV D-I
(3) A-I, IV B-II, III C-III, IV D-IV, III
(4) A-I, II B-II, III C-III, IV D-IV, I

41. The molar heat capacity for an ideal gas-

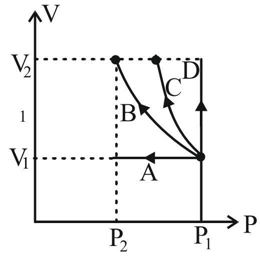
(a) is zero for adiabatic process.
(b) is infinite for an isothermal process
(c) depends on the nature of gas for a process in which either volume or pressure is constant.
(d) is equal to the product of the molecular weight and gram specific heat capacity for any process.

Choose the **correct** statements:-

(1) a, b
(2) c, d
(3) a, b, c & d
(4) b, c, d

42. Volume versus pressure curves for one mole of an ideal gas are given for four processes as shown in figure.

(B → Adiabatic process, C→ Isothermal process)



List-I		List-II	
(A)	For process A	(I)	Work done by the gas is positive.
(B)	For process B	(II)	Temperature will increase.
(C)	For process C	(III)	Heat supplied is positive.
(D)	For process D	(IV)	Change in internal energy is negative.

Choose the **correct** answer from the options given below:

(1) A-IV, B-I, IV, C-I, II D-I, II, III
(2) A-I, IV, B-I, C-I, III D-I, II, II
(3) A-I, IV B-IV C-I, III D-I, II
(4) A-IV, B-I, IV, C-I, III D-I, II, III

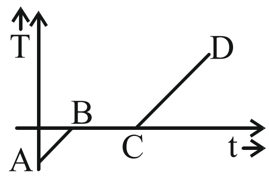
43. A good cooking utensil should have:

(A) low specific heat
(B) high specific heat
(C) low thermal capacity
(D) high coefficient of thermal conductivity.

Which statement are **correct**:

(1) A, C, D (2) B, C, D
(3) A, B, D (4) B, C

44. Heat is supplied to a ice at a constant rate temperature variation with time is a shown in figure, then-



(A) During AB volume of substance increases
(B) during BC volume of substance decreases
(C) Specific heat of substance in liquid phase is proportional to reciprocal of slope of protion AB of graph
(D) Latent heat of fusion of substance is independent of portion AB of graph.

The **correct** option is:-

(1) A, C (2) B, C
(3) A, C, D (4) B, D

45. Match the List with scale of temparature and its range:

List-I		List-II	
(A)	Celsius scale	(I)	32° to 212°
(B)	Fahrenheit scale	(II)	x° to y°
(C)	Kelvin scale	(III)	0° to 100°
(D)	Unknown scale	(IV)	273.15 to 373.15

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-III, D-IV
(2) A-III, B-I, C-IV, D-II
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-II, C-IV, D-I

46. A spherical black body of radius r radiates power P and its rate of cooling is R then-

(a) $P \propto r$ (b) $P \propto r^2$

(c) $R \propto r^2$ (d) $R \propto \frac{1}{r}$

Choice the **correct o**ne:-

(1) a, b (2) a, c
(3) b, d (4) a, b, d

47. Match the **List-I** and **List-II**:

List-I		**List-II**	
(A)	Heat	(I)	Degree of hotness or coolness of a substance
(B)	Internal energy	(II)	Amount of heat required to rise the temparature of unit mass of substance by 1°C
(C)	Temperature	(III)	Energy in transit
(D)	Specific heat	(IV)	The sum of energy due to attractive force between molecules and due to random motion

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-I, D-II
(2) A-III, B-I, C-IV, D-II
(3) A-III, B-II, C-I, D-IV
(4) A-III, B-IV, C-II, D-I

48. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: For higher temperature, the peak emission wavelength of a black body shifts to lower wavelengths.

Reason R: Peak emission wavelength of a black body is proportional to the fourth power of temperature.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

49. Match the **List-I** and **List-II**:

List-I		**List-II**	
(A)	Speed of sound	(I)	$\sqrt{\frac{2RT}{Mw}}$
(B)	RMS speed of gas molecule	(II)	$\sqrt{\frac{8}{\pi}\frac{RT}{Mw}}$
(C)	Average speed of gas molecule	(III)	$\sqrt{\frac{3RT}{Mw}}$
(D)	Most probable speed of gas molecule	(IV)	$\sqrt{\frac{\gamma RT}{Mw}}$

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-III, D-IV
(2) A-II, B-IV, C-I, D-III
(3) A-III, B-I, C-IV, D-II
(4) A-IV, B-III, C-II, D-I

50. Match the **List-I** and **List-II**:

List-I		**List-II**	
(A)	Adiabatic bulk modulus	(I)	$-P/V$
(B)	Slope of PV curve in isothermal process	(II)	$\frac{2}{\gamma-1}$
(C)	Degree of freedom	(III)	γP
(D)	Molar heat capacity at constant pressure divide by R	(IV)	$\frac{\gamma}{\gamma-1}$

Choose the **correct** answer from the options given below:

(1) A-III, B-I, C-II, D-IV
(2) A-I, B-III, C-IV, D-II
(3) A-II, B-IV, C-I, D-III
(4) A-IV, B-II, C-III, D-I

SECTION-(II) CHEMISTRY

SECTION - A

51. During electrolysis of an aqueous solution of sodium sulphate, 2.4 L of oxygen (O_2) at STP was liberated at anode, the volume of hydrogen (H_2) at STP, liberated at cathode would be:

(1) 1.2 L
(2) 2.4 L
(3) 2.6 L
(4) 4.8 L

52. If the E°_{cell} for a given reaction has a negative value, which of the following gives the **correct** relationships for the values of ΔG° and K_{eq}?

(1) $\Delta G^{\circ} < 0; K_{eq} > 1$
(2) $\Delta G^{\circ} < 0; K_{eq} < 1$
(3) $\Delta G^{\circ} > 0; K_{eq} < 1$
(4) $\Delta G^{\circ} > 0; K_{eq} > 1$

53. The IUPAC name of the complex $[Co(NH_3)_4(H_2O)Cl]Cl_2$ is:

(1) Aquatetraamminechloridocobalt (I) chloride
(2) Chloridoaquatetraamminechloridocobalt(II) chloride
(3) Chloridoaquatetraamminecobalt (IV) chloride
(4) Tetraammineaquachloridocobalt (III) chloride

54. When electric current is passed through acidified water, 112 mL of hydrogen gas at STP collected at the cathode in 965 seconds. The current passed in amperes is:

(1) 1.0 A
(2) 0.5 A
(3) 0.1 A
(4) 2.0 A

55. The resistance of 0.01 N solution of an electrolyte AB at 328 K is 100 ohm. The specific conductance of solution is: (cell constant = 1 cm^{-1})

(1) 100 ohm
(2) 10^{-2} ohm^{-1}
(3) 10^{-2} ohm^{-1} cm^{-1}
(4) 10^{2} ohm.cm

56. At 25°C, the standard emf of cell having reaction involving two electrons change is found to be 0.295 V. The equilibrium constant of the reaction is:

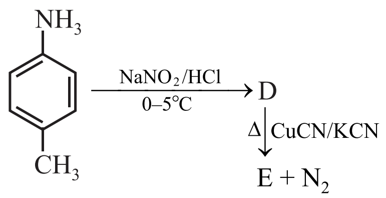
(1) 29.5×10^{-2}
(2) 10
(3) 10^{10}
(4) 29.5×10^{10}

57. Match **List-I** with **List-II:**

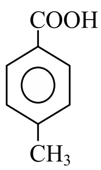
List-I (Common name)		List-II (IUPAC)	
(A)	Bromoform	(I)	Chlorophenylmethane
(B)	Vinyl chloride	(II)	Tribromomethane
(C)	Benzyl chloride	(III)	Chloroethene
(D)	Chloroform	(IV)	Trichloromethane

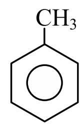
Choose the **correct** answer from the options give below:

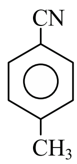
(1) A-III, B-II, C-I, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-III, C-I, D-IV
(4) A-II, B-I, C-III, D-IV

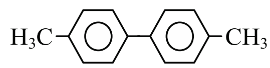
58. In the reaction 

The product E is:

(1) 

(2) 

(3) 

(4) 

59. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Reaction of alcohol with $SOCl_2$ gives alkyl chloride, sulphur dioxide and hydrogen chloride.

Reason R: In S_N1 reaction, the order of reactivity of alcohol is 1° > 2° > 3°.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

60. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

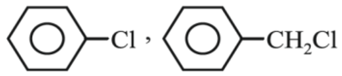
Assertion A: $[Fe(H_2O)_5NO]SO_4$ is paramagnetic.

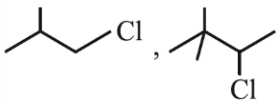
Reason R: The Fe in $[Fe(H_2O)_5NO]SO_4$ has three unpaired electrons.

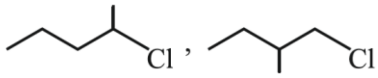
In the light of the above statements, choose the **correct** answer from the options given below:

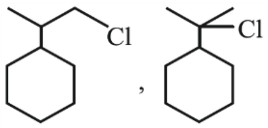
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

61. In which of the following pairs, the first compound is more reactive than second one for S_N1 reaction?

(1) 

(2) 

(3) 

(4) 

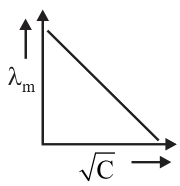
62. For lead storage battery, pick the **correct** statements.

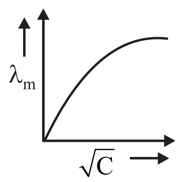
A. Lead storage battery has ~ 48% solution of sulphuric acid as an electrolyte.
B. During discharging of battery, Pb as anode is converted into $PbSO_4$.
C. Lead storage battery consists of grid of lead packed with PbO_2 as anode.
D. Lead storage battery has ~ 38% solution of sulphuric acid as an electrolyte.

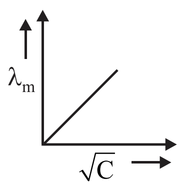
Choose the **correct** answer from the options given below:

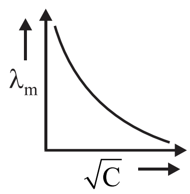
(1) B and D only
(2) B, C and D only
(3) A, B and D only
(4) B and C only

63. Which of the following curve represents the variation of λ_m with $\sqrt{C}$ for $AgNO_3$?

(1) 

(2) 

(3) 

(4) 

64. Which is **not** true about metal carbonyls?

(1) CO ligand acts as a Lewis base as well as Lewis acid.
(2) Metal acts as Lewis base as well as Lewis acid.
(3) $d\pi - p\pi$ back bonding takes place.
(4) $p\pi - p\pi$ back bonding takes place.

65. At 25°C, molar conductance of 0.1 M aqueous solution of ammonium hydroxide is 9.54 $ohm^{-1}\,cm^2\,mol^{-1}$ and at infinite dilution its molar conductance is 238 $ohm^{-1}\,cm^2\,mol^{-1}$. The degree of ionization of ammonium hydroxide at the same concentration and temperature is:

(1) 40.800% (2) 2.080%
(3) 20.800% (4) 4.008%

66. Effective atomic number of Fe in the complex $K_4[Fe(CN)_6]$ is:

(1) 24 (2) 12
(3) 36 (4) 18

67. Which of the following solution is used for the estimation of hardness of water?

(1) Hypo solution
(2) $KMnO_4$ solution
(3) Na_2EDTA solution
(4) $K_2Cr_2O_7$ solution

68. The ligand that shows synergic effect is:

(1) H_2O (2) NH_3
(3) CO (4) C_5H_5N

69. The most stable complex among the following is:

(1) $K_3[Al(C_2O_4)_3]$ (2) $[Pt(en)_2]Cl_2$
(3) $[Ag(NH_3)_2]Cl$ (4) $K_2[Ni(EDTA)]$

70. The value of electrode potential

$(10^{-4}\,M)\ H^+\,|\,H_2\ (1\ atm)\,|\,Pt$ at 298 K would be:

(1) –0.236 V (2) +0.404 V
(3) +0.236 V (4) –0.476 V

71. The specific conductance in $ohm^{-1}\,cm^{-1}$ of four electrolytes P, Q, R and S are given in brackets:

$P(5.0 \times 10^{-5})$ $Q(7.0 \times 10^{-8})$
$R(1.0 \times 10^{-10})$ $S(9.2 \times 10^{-3})$

The one that offers highest resistance to the passage of same amount electric current is:

(Cell constant = 1 cm^{-1})

(1) P (2) S
(3) R (4) Q

72. The compound in which the oxidation state of oxygen having –1 is:

(1) H_2O (2) O_2F_2
(3) Na_2O (4) BaO_2

73. Consider the following reaction:

A. $(CH_3)_3CCH(OH)CH_3 \xrightarrow{\text{conc. } H_2SO_4}$

B. $(CH_3)_2CHCH(Br)CH_3 \xrightarrow{\text{alc. KOH}}$

C. $(CH_3)_2CHCH(Br)CH_3 \xrightarrow{(CH_3)_3CO^-K^+}$

D. $(CH_3)_2CH-\underset{\displaystyle OH}{\underset{|}{C}H}-CHO \xrightarrow{\Delta}$

Which of these reaction(s) will **not** produce Saytzeff product?

(1) A, C and D only (2) D only

(3) C only (4) B and D only

74. Which step is chain propagation step in the following mechanism?

(1) $Cl_2 \xrightarrow{h\nu} \dot{C}l + \dot{C}l$

(2) $\dot{C}l + CH_4 \rightarrow \dot{C}H_3 + HCl$

(3) $\dot{C}l + \dot{C}l \rightarrow Cl_2$

(4) $\dot{C}H_3 + \dot{C}l \rightarrow CH_3Cl$

75. The **correct** order of reactivity of alkene towards an electrophile is:

(1) $CH_2 = CH-Cl > CH_2 = CH-OCH_3$

(2) $CH_2 = CHCl < CH_2 = CCl_2$

(3) $CH_2 = CH_2 > CH_3-CH = CH_2$

(4) $CH_2 = CH-OCH_3 > CH_2 = CH-CH_2-OH$

76. The oxidation states of iodine in HIO_4, H_3IO_5 and H_5IO_6 are respectively:

(1) +1, +3, +7 (2) +7, +7, +3

(3) +7, +7, +7 (4) +7, +5, +3

77. Which of the following expression is **not** correct?

(1) $\Delta G° = -nFE°$

(2) $\Delta G° = -RT \ln K_{eq}$

(3) $E° = -(RT/nF) \ln K_c$

(4) $\Delta G = \Delta G° + RT \ln Q$

78. In a reaction,

H_2O(steam) + C (glowing) $\rightarrow$ CO + H_2. Which of the following statement is **correct**?

(1) H_2O is the reducing agent.

(2) H_2O is the oxidising agent.

(3) carbon is the oxidising agent.

(4) oxidation-reduction does not occur.

79. Which of the following reactions involve disproportionation?

(1) $2H_2SO_4 + Cu \rightarrow CuSO_4 + 2H_2O + SO_2$

(2) $As_2O_3 + 3H_2S \rightarrow As_2S_3 + 3H_2O$

(3) $2KOH + Cl_2 \rightarrow KCl + KOCl + H_2O$

(4) $Ca_3P_2 + 6H_2O \rightarrow 3Ca(OH)_2 + 2PH_3$

80. Only two isomeric monochloro derivatives are possible for:

(1) 2-Methylpropane

(2) n-Pentane

(3) Benzene

(4) 2,4-Dimethylpentane

81. $H_3C-\underset{\displaystyle CH_3}{\underset{|}{C}H}-CH=CH_2 + HBr \rightarrow A$

A (predominantly) is:

(1) $CH_3-\underset{\displaystyle Br}{\underset{|}{C}H}-\underset{\displaystyle CH_3}{\underset{|}{C}H}-CH_3$

(2) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{C}H}-\underset{\displaystyle Br}{\underset{|}{C}H}-CH_3$

(3) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{C}H}-CH_2-CH_2Br$

(4) $CH_3-\overset{\displaystyle Br}{\overset{|}{\underset{\displaystyle CH_3}{\underset{|}{C}}}}-CH_2CH_3$

82. Sodalime decarboxylation of sodium propionate produces:

(1) Propane.

(2) Ethane.

(3) Methane.

(4) Butane.

83. What is the oxidation state of S in Na_2S_2?

(1) +1

(2) –2

(3) –1

(4) 0

84. Given below are two statements:

Statement I: The melting point of neopentane is greater than that of n-pentane but the boiling point of n-pentane is more than that of neopentane.

Statement II: Melting point depends upon packing in crystal lattice whereas boiling point depends upon the surface area of the molecule.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

85. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Fe reacts with HCl to produce H_2 gas.

Reason R: Fe is a better reducing agent than H_2.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is not the correct explanation of A.

SECTION-B

86. Match the **List-I** with **List-II:**

List-I (Ions)		**List-II (Oxidation states)**	
(A)	$Cr_2O_7^{2-}$	(I)	+3
(B)	MnO_4^-	(II)	+5
(C)	VO_3^-	(III)	+6
(D)	FeF_6^{3-}	(IV)	+7

Choose the **correct** answer from the options give below:

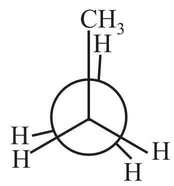
(1) A-III, B-IV, C-II, D-I

(2) A-I, B-IV, C-II, D-III

(3) A-III, B-II C-IV, D-I

(4) A-III, B-IV, C-I, D-II

87. Following eclipsed form of propane is repeated after rotation of:



(1) 45°

(2) 90°

(3) 120°

(4) 180°

88. Match **List-I** with **List-II**.

List-I		**List-II**	
(A)	Oxidising agent	(I)	Disproportionation
(B)	Reducing agent	(II)	Redox reaction
(C)	$2Cu^+ \rightarrow Cu^{2+} + Cu$	(III)	Decreases its oxidation number
(D)	$H_2O_2 + O_3 \rightarrow H_2O + 2O_2$	(IV)	Increases its oxidation number

Choose the **correct** answer from the options give below:

(1) A-III, B-IV, C-I, D-II

(2) A-IV, B-III, C-I, D-II

(3) A-III, B-IV, C-II, D-I

(4) A-II, B-I, C-III, D-IV

89. Given below are two statements:

Statement I: 2-Methylbutane on oxidation with $KMnO_4$ gives 2-methylbutan-2-ol.

Statement II: n-Alkanes can be easily oxidised to corresponding alcohols with $KMnO_4$.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

90. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: When 1 M $CuSO_4$(aq) solution is electrolysed using copper electrodes, copper is dissolved at anode and copper gets deposited at cathode.

Reason R: The standard oxidation potential of copper is less than the standard oxidation potential of water and standard reduction potential of copper is greater than the standard reduction potential of water.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

91. Given, $E^\circ_{Fe^{3+}/Fe} = -0.036\,V, E^\circ_{Fe^{2+}/Fe} = -0.439\,V$

The value of standard electrode potential for the change,

$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$ will be:

(1) 0.385 V
(2) 0.770 V
(3) –0. 270 V
(4) –0.072 V

92. When acetylene is passed through dil. H_2SO_4 in the presence of $HgSO_4$, the compound formed is:

(1) C_2H_5OH (2) CH_3COCH_3
(3) CH_3CHO (4) CH_3CH_3

93. The most stable complex ion:

(1) $[Fe(OH)_6]^{3-}$
(2) $[Fe(Cl)_6]^{3-}$
(3) $[Fe(CN)_6]^{3-}$
(4) $[Fe(H_2O)_6]^{3+}$

94. Resistance of a decimolar solution between two electrodes 0.02 m apart and 0.004 m^2 in area was found to be 50 ohm. The molar conductance of solution is:

(1) 0.1 S $cm^2\,mol^{-1}$
(2) 10 S $cm^2\,mol^{-1}$
(3) 10^{-3} S $cm^2\,mol^{-1}$
(4) 10^3 S $cm^2\,mol^{-1}$

95. $Pt(s)\,|\,H_2(g, 1\text{ bar})\,|\,H^+(aq, 1\text{ M})\,||\,Cu^{2+}(aq, 1\text{ M})\,|\,Cu$

The measured emf of the cell is 0.34 V means:

(1) Cu^{2+} ions get reduced easily than H^+ ion.
(2) hydrogen ions cannot oxidise Cu.
(3) copper does not dissolve in HCl.
(4) All of these

96. In the reaction, $C_2H_5OH \xrightarrow{(x)} C_2H_5Cl$,

Reagent (x) will be:

(1) NaCl (2) PCl_5
(3) Cl_2 (4) KCl

97. 100 mL of 0.3 M Fe^{3+}(aq) solution were electrolysed by a charge of 0.072 F. In electrolysis, metal was deposited and O_2(g) was evolved. At the end of electrolysis, it is desired to oxidize the unelectrolysed metal ion.

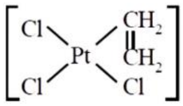
$Fe^{3+} + e^- \rightarrow Fe^{2+}$

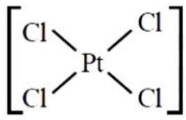
$Fe^{2+} + 2e^- \rightarrow Fe$

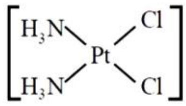
The moles of Fe^{2+} ions left unelectrolysed in the solution is:

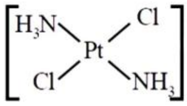
(1) 0.009 (2) 0.021
(3) 0.072 (4) 0.042

98. Which of the following is considered to be anticancer species?

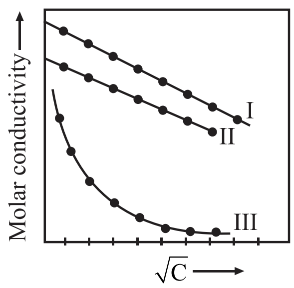
(1) 

(2) 

(3) 

(4) 

99. A graph was plotted between molar conductivity of various electrolytes (NaCl, HCl and NH_4OH) and $\sqrt{C}$ (in mol L^{-1}). The **correct** set is:



(1) I-(NaCl), II-(HCl), III-(NH_4OH)
(2) I-(HCl), II-(NaCl), III-(NH_4OH)
(3) I-($(NH)_4OH$), II-(NaCl), III-(HCl)
(4) I-(NH_4OH), II-(HCl), III-(NaCl)

100. The degenerate orbitals of $[Cr(H_2O)_6]^{3+}$ are:

(1) d_{yz} and d_{z^2}
(2) d_{z^2} and d_{xz}
(3) d_{xz} and d_{yz}
(4) $d_{x^2-y^2}$ and d_{xy}

SECTION-(III) BOTANY

SECTION - A

101. The **correct** equation of expressing the geometric growth is: [W_1 = Final size, W_0 = Initial size]
(1) $W_0 = W_1 e^{rt}$.
(2) $W_1 = W_0 e^{rt}$.
(3) $W_0 = W_t + rt$.
(4) $W_t = W_0 + rt$.

102. Identify from the following which is/are related to monocot leaf.
A. In grasses, certain adaxial epidermal cells along the veins modify themselves into large and empty cells. These are called bulliform cells.
B. Bulliform cells in the leaves make the leaf surface exposed.
C. Bulliform cells in the leaves make the leaves curl inwards.
D. Well differentiated mesophyll.
Choose the most appropriate answer from the options given below:
(1) Only A, C and D are correct
(2) Only A, B and C are incorrect
(3) Only A, B and C are correct
(4) Only B and D are correct

103. Select the **correct** aestivation and its example for the figure showing the position of floral parts on thalamus.

(1) Hypogynous in brinjal
(2) Hypogynous in peach
(3) Perigynous in rose
(4) Epigynous in rayflorets of sunflower

104. Select the **incorrect** statement w.r.t cells of meristematic zone during growth.
(1) The cells are rich in protoplasm.
(2) Possess small nucleus.
(3) Cell walls are primary in nature.
(4) Abundant plasmodesmatal connection.

105. Given below are two statements:
Statement I: Auxin is called plant growth promoter.
Statement II: Auxins promote the abscission of young leaves and fruits.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

106. Select the **incorrect** option from the following.
(1) E. Kurosawa (1926) reported the appearance of symptoms of the bakanae disease in canary grass seedlings.
(2) Auxin was isolated by F.W. Went from tips of coleoptiles of oat seedlings.
(3) The PGRs can be broadly divided into two groups based on their functions in a living plant body.
(4) Dormin is ABA.

107. Which of the following reaction in Kreb cycle will **not** yield NADH?
A. Oxaloacetic acid to Citric acid
B. Succinic acid to malic acid
C. Succinyl CoA to succinic acid
D. Citric acid to Succinic acid
Choose the most appropriate answer from the options given below:
(1) A and C only
(2) B and D only
(3) A and D only
(4) A, B and C only

108. Which of the following statement is **incorrect** with respect to roots?
(1) In monocotyledonous plants, the primary root is short lived and is replaced by a large number of roots.
(2) In some plants, roots arise from parts of the plant other than the radicle and are called fibrous roots.
(3) In majority of the dicotyledonous plants, the direct elongation of the radicle leads to the formation of primary root which grows inside the soil.
(4) *Monstera* and the banyan tree have adventitious roots.

109. Select the **mismatched** pair for placentation.

(1)	Axile	Lemon
(2)	Basal	Marigold
(3)	Free central	Primrose
(4)	Parietal	Tomato

110. Match **List-I** with **List-II**:

List-I		**List-II**	
(A)	Cuticle	(I)	Pore
(B)	Trichomes	(II)	Unicellular
(C)	Root hair	(III)	Waxy layer
(D)	Stomata	(IV)	Usually multicellular

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-II, D-I
(4) A-II, B-I, C-III, D-IV

111. Given below are two statements:

Statement I: The biomolecule which is common to a respiration-mediated breakdown of fats, carbohydrates and proteins is pyruvic acid.

Statement II: Pyruvic acid, the key product of glycolysis can have many metabolic fates.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

112. Leaf base if becomes swollen, is termed as __**A**__ and is found in __**B**__. Identify **A** and **B** respectively.

(1) A-receptacle, B-cereals
(2) A-receptacle, B-legumes
(3) A-pulvinus, B-cereals
(4) A-pulvinus, B-legumes

113. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The outermost layer of dicotyledonous root is epiblema.

Reason R: All cells of epiblema protrude in the form of root hairs.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

114. Select which are **not** present in roots.

A. Cuticle B. Mesophyll
C. Stele D. Conjuctive tissue

Choose the most appropriate answer from the options given below:

(1) A and B only (2) B and D only
(3) A and D only (4) B and C only

115. Select the **correct** match for sorghum from the following.

A. Mesophyll cell – RuBisCo
B. Bundle sheath – Decarboxylation
C. Bundle sheath – PEP carboxylase
D. Mesophyll cell – Oxaloaceticacid

Choose the most appropriate answer from the options given below:

(1) A and C only (2) B and D only
(3) A and D only (4) B and C only

116. Select the **incorrect** match.

(1)	Radial vascular bundle	Monocot root
(2)	Polyarch xylem	Dicot root
(3)	Well-developed pith	Dicot stem
(4)	Well-developed pith	Monocot root

117. The first stable product of Calvin cycle has:

(1) 2 carbon atoms and 3 phosphate group.
(2) 3 carbon atoms and 1 phosphate group.
(3) 3 carbon atoms and 2 phosphate group.
(4) 1 carbon atoms and 3 phosphate group.

118. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Each flower normally has five floral whorls.

Reason R: Gulmohar is zygomorphic.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

119. Identify the different types of aestivations (**A, B** and **C**) in corolla and select the **correct** option.

(A) (B) (C)

(1) A-Imbricate, B-Twisted and C-Valvate
(2) A-Vexillary, B-Valvate and C-Twisted
(3) A-Imbricate, B-Twisted and C-Vexillary
(4) A-Valvate, B-Twisted and C-Imbricate

120. Select the **incorrect** option for epidermis.
(1) Outermost layer of the primary plant body.
(2) It is made up of elongated cell.
(3) Epidermal cells have abundant cytoplasm.
(4) Epidermis forms a continuous layer.

121. Difference in shapes of leaves produced in air and those produced in water is found in:
(1) cotton, coriander and larkspur.
(2) cotton, coriander and buttercup.
(3) buttercup, coriander and larkspur.
(4) buttercup only.

122. Read the following statements for fruit.
A. Fruit is a mature or ripened ovary always developed before fertilisation.
B. It consists of pericarp also known as seed.
C. When fruit wall is thick and fleshy, it is differentiated into outer mesocarp, middle epicarp and inner endocarp.
D. In mango and coconut, the fruit is known as a berry.
Choose the most appropriate answer from the options given below:
(1) Only A and C are correct
(2) A, B, C and D are incorrect
(3) Only A, B and C are correct
(4) Only B and D are correct

123. Plant tissue system include/s:
(1) epidermal tissue system and vascular tissue system only.
(2) ground tissue and epidermal tissue system only.
(3) ground and vascular tissue system only.
(4) epidermal, fundamental and conducting tissue system.

124. Respiratory quotient may be represented as:
(1) O_2 taken in/CO_2 evolved.
(2) CO_2 evolved/O_2 taken in.
(3) CO_2 taken in/O_2 evolved.
(4) O_2 evolved/CO_2 taken in.

125. Which of the given is **not** formed in oxidative decarboxylation?
(1) Acetyl CoA (2) CO_2
(3) NADH + H^+ (4) NADPH + H^+

126. Select the **incorrect** for alcohol fermentation.
A. CO_2 is formed.
B. Yeasts poison themselves to death when the concentration of alcohol reaches about 13 percent.
C. 2 ATP are generated.
D. NADH + H^+ is formed.
Choose the most appropriate answer from the options given below:
(1) A, B and C only (2) B and D only
(3) D only (4) B and C only

127. Match **List-I** with **List-II:**

List-I		List-II	
(A)	Monoadelphous	(I)	Pea
(B)	Polyadelphous	(II)	China rose
(C)	Diadelphous	(III)	*Citrus*
(D)	Epipetalous	(IV)	Brinjal

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-II, D-I
(4) A-II, B-I, C-III, D-IV

128. Select the **incorrect** statements regarding gibberellins.
A. GA_1 is the most intensively studied form.
B. Gibberellins cause apple to elongate.
C. Gibberellins promotes bolting in beet.
D. Gibberellic acid (GA_3) was one of the first gibberellins to be discovered.
Choose the most appropriate answer from the options given below:
(1) A only (2) B and D only
(3) A and D only (4) B and C only

129. Bracteate in flower is:
(1) reduced sepal found at the base of the thalamus.
(2) reduced leaf found at the base of the pedicel.
(3) reduced androecium found at the base of the leaf.
(4) reduced ovary found at the base of the pulvinus.

130. The F_0 and F_1 headpiece is/are:
(1) integral membrane protein complex and peripheral membrane protein complex respectively.
(2) peripheral membrane protein complex and integral membrane protein complex respectively.
(3) integral membrane protein complex.
(4) peripheral membrane protein complex.

131. C_4 plants show saturation of CO_2 at about _______ μlL^{-1}.
(1) 360 (2) 450
(3) 110 (4) 200

132. CO_2 is required for photosynthesis was proved by all **except**:
(1) half leaf exposed to air and half in test tube.
(2) KOH soaked cotton placed near the test tube.
(3) starch test.
(4) the portion that was in the tube, tested negative.

133. Given below are two statements:

Statement I: 18 ATP are used to form one glucose molecule in C_3 cycle.

Statement II: Stroma lamellae membranes lack PS II as well as NADP reductase enzyme.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

134. Cortex consists of three sub-zones in dicot stem, they are:

I. Hypodermis
II. Cortical layer
III. Pericycle
IV. Endodermis
V. Epidermis

(1) All except I and II
(2) All except II and III
(3) All except IV and V
(4) All except III and V

135. Read the following and select **incorrect** statement.

(1) Photochemical phase include the formation of high-energy chemical intermediates.
(2) The LHC are made up of hundreds of pigment molecules bound to vitamins.
(3) Photosystem are named in the sequence of their discovery
(4) The reaction centre is different in both the photosystems.

SECTION - B

136. Photorespiration is a wasteful process because:

(1) there is synthesis of ATP only.
(2) there is utilisation of NADPH only.
(3) synthesis of phosphoglycolate(3C) molecule.
(4) there is no synthesis of sugars.

137. RuBisCO enzyme in plants has ________ when CO_2:O_2 is nearly equal.

(1) more affinity for CO_2, than for O_2.
(2) more affinity for O_2, than for CO_2.
(3) equal affinity for both.
(4) more affinity for sugars, than for CO_2.

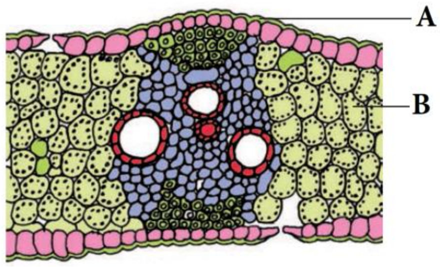
138. Match **List-I** with **List-II:**

List-I		List-II	
(A)	Oxidative decarboxylation	(I)	Citrate synthase
(B)	Glycolysis	(II)	Pyruvate dehydrogenase
(C)	Oxidative phosphorylation	(III)	Electron transport system
(D)	Tricarboxylic acid cycle	(IV)	EMP pathway

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-II, D-I
(4) A-II, B-IV, C-III, D-I

139. Transverse section of isobilateral leaf passing through the midrib is given below. Identify **A** and **B** and select the **correct** labelling for **A** and **B**.



(1) A-Adaxial epidermis, B-Spongy mesophyll
(2) A-Abaxial epidermis, B-Palisade mesophyll
(3) A-Adaxial epidermis, B-Mesophyll
(4) A-Abaxial epidermis, B-Mesophyll

140. Which of the following statement is **incorrect** regarding Z-scheme?

(1) Photosystem-I receives electrons from photosystem-II.
(2) Photosystem-II receives electrons from photolytic dissociation of water.
(3) Formation of NADPH is associated with photosystem-II.
(4) Reaction centre of photosystem I is P700.

141. There is a linear relationship between incident light and CO_2 fixation rates at __________ light intensities.

(1) high
(2) low
(3) moderate
(4) very high

142. Select how many is/are **incorrect** for pigment colour in chromatogram.

A. Chlorophyll *a* - Yellow green

B. Chlorophyll *b* - Bright or blue green

C. Xanthophylls - Yellow to yellow-orange

D. Carotenoids - Yellow

Choose the most appropriate answer from the options given below:

(1) One (2) Two

(3) Three (4) Four

143. During the process of respiration, which of the followings are released as products?

(1) CO_2, H_2O and O_2.

(2) CO_2, H_2O and energy.

(3) O_2, H_2O and energy.

(4) CO, H_2O and energy.

144. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Glycolysis can use fructose also as substrate.

Reason R: Glucose and fructose are phosphorylated to give rise to glucose-6- phosphate by the activity of the enzyme hexokinase.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is not the correct explanation of A.

145. Select **correct** statement for glycolysis.

(1) Glyceraldehyde-3-phosphate to 1,3 bisphosphoglyceric acid conversion yield NADH + H^+.

(2) Glycolysis occurs in the cytoplasm of the mitochondria.

(3) One molecule of pyruvic acid is formed during glycolysis.

(4) The term glycolysis has originated from the Latin words, *glycos* for sugar, and *lysis* for splitting.

146. Match **List-I** with **List-II**:

	List-I		List-II
(A)	Auxin	(I)	Early seed production in conifers
(B)	Abscisic acid	(II)	Controls the xylem differentiation
(C)	Ethylene	(III)	Closure of stomata
(D)	Gibberellin	(IV)	Female flowers in cucumbers

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I, D-IV

(2) A-II, B-III, C-IV, D-I

(3) A-III, B-IV, C-II, D-I

(4) A-II, B-I, C-III, D-IV

147. Given below are two statements:

Statement I: T.W Engelmann using a prism split light into its spectral components.

Statement II: The bacteria were used to detect the sites of CO_2 evolution.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

148. The enzymes of dark reaction in C_4 plants are found in the:

A. thylakoid.

B. stroma of chloroplast of mesophyll cells.

C. stroma of chloroplast of bundle sheath cells.

D. stromal lamella.

Choose the most appropriate answer from the options given below:

(1) A and C only

(2) B and D only

(3) A and D only

(4) B and C only

149. Which of the following represents the **correct** sequence of the development process in a plant cell?

(1) Senescence → Elongation → Cell division → Maturation

(2) Meristematic cell → Senescence → elongation → Death

(3) Elongation → Senescence → Maturation → Plasmatic growth

(4) Meristematic cell → Differentiation → Maturation → Senescence

150. Relative growth rate which is also referred to as efficiency index is:

(1) also the measure of the ability of the plant to produce new plant material.

(2) increase in number of cell.

(3) increase in number of weight.

(4) arithmetic growth rate.

SECTION-(IV) ZOOLOGY

SECTION - A

151. How many of the following parts are included in the hindbrain?

Pons, Thalamus, Corpus callosum, Hypothalamus, Cerebrum, Medulla

(1) Four
(2) Five
(3) Three
(4) Two

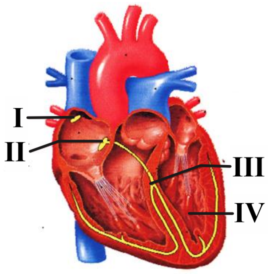
152. Which of the following can be considered as point of difference between cortical and juxtamedullary nephrons?
(1) Their number.
(2) Length of loop of Henle.
(3) Presence of *vasa recta*.
(4) All of these

153. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In amphibians and reptiles, the left atrium receives oxygenated blood and right atrium receives deoxygenated blood.
Reason R: The oxygenated blood gets mixed up with deoxygenated blood in double ventricles in amphibians and reptiles.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

154. How many filaments of tropomyosin run close to the F actins throughout its length?
(1) Ten
(2) Two
(3) Six
(4) Eight

155. Depolarisation of axonal membrane is caused due to:
(1) influx of Na^+ ions.
(2) efflux of Na^+ ions.
(3) influx of K^+ ions.
(4) efflux of K^+ ions.

156. The diagram given below shows a section of a human heart with labelled **I**, **II**, **III** and **IV**.
Select the **correct** option which gives **correct** identification.



(1)	I	Atrio-ventricular node is present in the right lower corner of the left atrium.
(2)	II	Sino-atrial node present in the upper left corner of the left atrium.
(3)	III	Bundle of His transmits action potential to the ventricles.
(4)	IV	Thin -walled, the interatrial septum separates the left and the right ventricles.

157. Which of the following hormones inhibits gastric secretion and motility?
(1) Gastrin
(2) Gastric inhibitory peptide
(3) Atrial natriuretic factor
(4) Thymosin

158. Deficiency of albumin proteins in the blood plasma would affect the:
(1) osmotic balance of body.
(2) defense mechanisms of the body.
(3) coagulation of blood.
(4) transport of gases by blood.

159. Given below are two statements:
Statement I: Prolonged hyperglycemia leads to a complex disorder called diabetes mellitus.
Statement II: Ovary is the secondary female sex organ that produces one ovum during each menstrual cycle.
In the light of the above statements, choose the most appropriate answer from the options given below
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

160. Which of the following hormones is secreted by the non-endocrine tissues in our body?
(1) Growth factors
(2) Catecholamines
(3) Thyrocalcitonin
(4) Vasopressin

161. Given below are two statements:
Statement I: Each branch of the axon terminates as a bulb-like structure called synaptic knob.
Statement II: Multipolar neurons are found in the cerebral cortex of the brain.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

162. Match **List-I** with **List-II** w.r.t hormone and its source glands.

List-I		List-II	
(A)	Melanocyte stimulating hormone	(I)	Thyroid gland
(B)	Thyroid stimulating hormone	(II)	Pars intermedia
(C)	Thyroxine	(III)	Adrenal gland
(D)	Corticoids	(IV)	Pars distalis

Choose the **correct** answer from the options given below:
(1) A-I, B-IV, C-II, D-III
(2) A-II, B-I, C-IV, D-III
(3) A-II, B-IV, C-I, D-III
(4) A-IV, B-III, C-I, D-II

163. Which of the following is **not** a function of parathyroid hormone (PTH)?
(1) PTH stimulates the process of bone resorption.
(2) PTH stimulates reabsorption of Ca^{2+} by the renal tubules.
(3) PTH increases Ca^{2+} absorption from the digested food.
(4) PTH lowers the blood Ca^{2+} levels.

164. Which of the following options is **correct** about the first heart sound?
(1) Dub sound which is heard due to the closure of the bicuspid valve.
(2) Lub sound which is heard due to the closure of the atrio-ventricular valves.
(3) Dub sound which is heard due to the closure of semilunar valves.
(4) Lub sound which is heard due to the closure of semilunar valves.

165. Which of the following options is **correct** w.r.t the nodes of Ranvier?
(1) These are present on dendrites of neurons.
(2) These are interruptions between myelin sheaths on an axon.
(3) They do not play any role in impulse conduction across the neuron.
(4) They are located in unmyelinated nerve fibres.

166. Identify the disease based on the given characteristics and choose the **correct** option.
I. It is caused due to less secretion of antidiuretic hormone.
II. Kidneys lose the ability to conserve water.
III. More water loss and dehydration occurs in this disease.
(1) Diabetes mellitus
(2) Exophthalmic goitre
(3) Diabetes insipidus
(4) Acromegaly

167. Read the following statements **(I-V).**
I. By releasing the ADP and P_i, the myosin goes back to its relaxed state.
II. Repeated activation of the skeletal muscles can lead to the accumulation of lactic acid.
III. Skeletal muscle contraction is initiated by a signal sent by the autonomic nervous system (ANS).
IV. The skeletal muscle bundles are held together by an epithelial tissue layer called fascia.
V. Muscles are endodermally derived.
Which of the above statements are **incorrect**?
(1) I and II only
(2) II, III and IV only
(3) III, IV and V only
(4) I, III and V only

168. Complete the analogy w.r.t. each human limb and choose the **correct** option.
Carpals: 8 : : Tarsals : ___________
(1) 16 (2) 14
(3) 8 (4) 7

169. Match **List-I** with **List-II**.

List-I		**List-II**	
(A)	Electrical synapse	(I)	Transmit impulses away from the cell body.
(B)	Afferent nerve fibres	(II)	Retina of eyes
(C)	Bipolar neurons	(III)	Transmit impulses from tissues/organs to the CNS.
(D)	Axon	(IV)	Rare in human system

Choose the **correct** answer from the options given below:

(1) A-I, B-IV, C-II, D-III

(2) A-II, B-I, C-IV, D-III

(3) A-III, B-I, C-II, D-IV

(4) A-IV, B-III, C-II, D-I

170. Which of the following sets of hormones have membrane bound receptors?

(1) Follicle stimulating hormone

(2) Androgen

(3) Epinephrine

(4) Both (1) and (3)

171. Which of the following pairs is **incorrectly** matched?

(1) Eosinophils : Allergic reactions

(2) Basophils : Inflammatory reactions

(3) Monocytes: Phagocytic action

(4) Neutrophils : Secrete histamines and serotonin

172. Match **List-I** with **List-II** w.r.t excretory organs.

List-I		**List-II**	
(A)	Flatworm	(I)	Kidney
(B)	Prawn	(II)	Green gland
(C)	*Periplaneta*	(III)	Flame cell
(D)	Mammals	(IV)	Malpighian tubule

Choose the **correct** answer from the options given below:

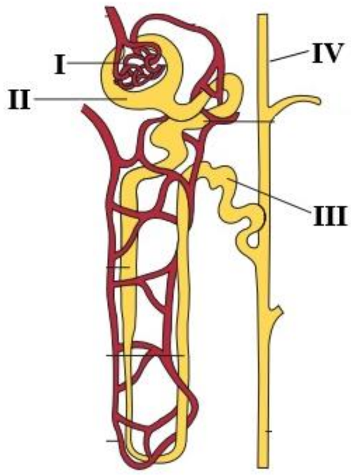
(1) A-III, B-I, C-IV, D-II

(2) A-II, B-I, C-IV, D-III

(3) A-III, B-II, C-IV, D-I

(4) A-IV, B-II, C-III, D-I

173. The given below diagram is the representation of a nephron with labelled **I**, **II**, **III** and **IV**. Identify the part where filtrate in nephron is called urine.



(1) I (2) II

(3) III (4) IV

174. Which of the following will **not** cause micturition?

(1) Stretching of urinary bladder.

(2) Contraction of smooth muscle of bladder.

(3) Relaxation of muscles of sphincter.

(4) Activation of stretch receptor in the wall of ureters.

175. Read the following statements **(A-D)** and choose the **correct** option.

A. Bowman's capsule is double walled cup-like structure.

B. Glomerulus along with Bowman's capsule is called the Malpighian body.

C. Elimination of urea requires the least amount of water.

D. Urine formation involves filtration, reabsorption and secretion.

Choose the most appropriate answer from the options given below:

(1) B, C and D only

(2) A, B and D only

(3) A, B and C only

(4) A, B, C and D

176. Choose the option **correctly** w.r.t the function of somatostatin hormone.

(1) To stimulate the release of growth hormone from the anterior pituitary gland.

(2) To inhibit the release of growth hormone from the anterior pituitary gland.

(3) To stimulate the release of gonadotrophins from the posterior pituitary gland.

(4) To inhibit the release of gonadotrophins from the posterior pituitary gland.

177. During cardiac cycle the flow of blood into the ventricles increases by about 30 per cent during:
(1) atrial systole. (2) ventricular systole.
(3) joint diastole. (4) atrial diastole.

178. The hormone called glucagon is secreted by:
(1) β-cells of Islet of Langerhans of pancreas.
(2) cells of adrenal gland.
(3) cells of parathyroid gland.
(4) α-cells of Islet of Langerhans of pancreas.

179. Inflammation of the glomeruli of the kidney referred to:
(1) cystitis.
(2) nephritis.
(3) glomerulonephritis.
(4) uremia.

180. On an average, ____**X**____ ml of blood is filtered by the kidneys per minute which constitute roughly ____**Y**____ of the blood pumped out by each ventricle of the heart in a minute.
Choose the option which fill the blanks **correctly**.

	X	**Y**
(1)	500–600	$1/5^{th}$
(2)	1100–1200	$1/3^{rd}$
(3)	500–600	$1/3^{rd}$
(4)	1100–1200	$1/5^{th}$

181. Which of the following pairs is **correctly** matched?

(1)	Hinge joint	Between vertebrae
(2)	Pivot joint	Between carpal and metacarpal of thumb
(3)	Cartilaginous joint	Between carpals
(4)	Fibrous joint	Flat skull bones

182. Which of the following features is common in both fish and humans?
(1) Presence of pulmonary circulation.
(2) Numbers of chambers of heart.
(3) Heart pumps only deoxygenated blood.
(4) Presence of closed circulatory system.

183. During skeletal muscle contraction, which of the following events occur?
I. H-zone disappears.
II. A-band widens.
III. I-band reduces in length.
IV. M-line and Z-line come closer.
Choose the **correct** answer from the options given below:
(1) I, III and IV only
(2) I, II and IV only
(3) II and IV only
(4) I, II and III only

184. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): Tropomyosin runs along the length of the actin filament, interacting with the F-actin filaments.
Reason (R): Tropomyosin is made up of monomeric proteins called meromyosins.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is not the correct explanation of A.

185. Read the following statements (**I-V**).
I. Our vertebral column is formed by 26 serially arranged units called vertebrae and is dorsally placed.
II. Smooth muscles activities are not under the voluntary control of the nervous system.
III. In the centre of each 'I' band is an elastic fibre called 'Z' line which bisects it.
IV. H zone is the central part of thick filament, not overlapped by thin filaments.
V. There are two pairs of floating ribs in the human body.
Which of the above statements are **correct**?
(1) I, II and V only
(2) I, III and IV only
(3) I, II, III and IV only
(4) I, II, III, IV and V

SECTION - B

186. Match **List I** with **List II** w.r.t total number of bones in human beings.

List I		**List II**	
(A)	Sternum	(I)	Eight
(B)	Femur	(II)	Ten
(C)	Metatarsals	(III)	Two
(D)	Cranial bones	(IV)	One

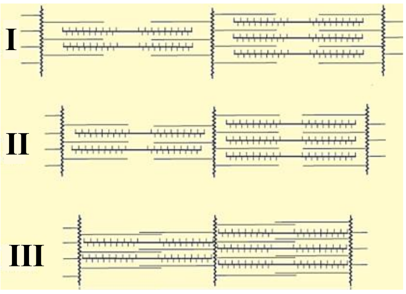
Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-II, D-I
(2) A-II, B-I, C-IV, D-III
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-I, C-IV, D-II

187. Which of the following functions are majorly performed by the association areas of the human brain?

A. Memory
B. Regulation of body temperature
C. Communication
D. Expression of emotional reactions

(1) A and B only
(2) B and C only
(3) C and D only
(4) A and C only

188. Consider the given below diagram and choose the **correct** option.



(1) The pumping of Ca^{++} ions back to the sarcoplasmic cisternae, resulting in the unmasking of actin filaments, leads to 'I'.
(2) 'II' leads to the return of 'Z' lines back to their original position.
(3) 'III' leads to the outward movement of the I band.
(4) 'III' represents muscular dystrophy.

189. Which of the following sets of hormones can easily pass through the cell membrane of target cell and bind to specific intracellular receptors?

I. Progesterone
II. Aldosterone
III. Estrogen
IV. Thyroxine

(1) II and III only
(2) I, II and III only
(3) I and III only
(4) I, II, III and IV

190. Identify the set of **incorrect** statements from the following.

A. The hypothalamus is the basal part of diencephalon, forebrain.
B. Exophthalmic goitre is caused due to hyperthyroidism.
C. Thyroxines do not play any role in the regulation of the basal metabolic rate.
D. Prolactin plays a very important role in the regulation of a 24-hour rhythm of our body.

(1) A and B only
(2) B and C only
(3) C and D only
(4) A and D only

191. Which of the following bones articulates with the occipital condyles to form a dicondylic skull?

(1) Atlas
(2) Axis
(3) Hyoid bone
(4) Frontal bone

192. Given below are two statements:

Statement I: Atrial natriuretic factor (ANF) can cause vasodilation and thereby decrease the blood pressure.

Statement II: Conditional reabsorption of Na^+ and water takes place in the distal convoluted tubule of the renal tubule.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

193. Which of the following hormones are produced by the pars distalis region of the pituitary gland?

A. Prolactin
B. Oxytocin
C. Growth hormone
D. Vasopressin
E. Luteinising hormone

(1) A, B and C only
(2) A, C and E only
(3) B, D and E only
(4) B, C and D only

194. Which of the following statements is **incorrect**?

(1) Neurons are excitable cells.
(2) Diffusion of K^+ outside the membrane helps to restore the resting potential of the membrane at the site of excitation.
(3) The axonal membrane is comparatively more permeable to sodium ions and nearly impermeable to potassium ions during resting state.
(4) Unipolar neurons has cell body with one axon only.

195. Given below are two statements:

Statement I: Human kidneys can produce urine nearly ten times concentrated than the initial filtrate formed.

Statement II: Reabsorption of water occurs actively in the initial segments of the nephron.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect

196. Angiotensin-II is responsible for all of the following, **except:**

(1) vasoconstriction.

(2) increase in blood pressure.

(3) release of aldosterone.

(4) decrease reabsorption of Na^+ from renal tubule.

197. Which of the following is a **correct** match for disease and its symptoms?

(1)	Tetany	High Ca^{2+} level causing rapid spasms.
(2)	Myasthenia gravis	Genetic disorder resulting in weakening and paralysis of smooth muscle.
(3)	Muscular dystrophy	An auto-immune disorder causing progressive degeneration of skeletal muscle.
(4)	Arthritis	Inflammation of joints.

198. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Unmyelinated nerve fibre is enclosed by a Schwann cell.

Reason R: Schwann cells form a myelin sheath around the cell body of myelinated nerve fibres.

In the light of the above statements, choose the correct answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is a correct explanation of A.

(4) Both A and R are true but R is not a correct explanation of A.

199. Protein makes 6-8 percent of plasma which is more to percent of __**X**_____ of total leucocyte.

Choose the option which fills the blank **correctly**.

(1) eosinophils (2) neutrophils

(3) monocytes (4) lymphocytes

200. Read the following statements **(A-E)** and select the **incorrect** statements.

A. O blood group can be donated to persons with any other blood group.

B. Erythroblastosis foetalis condition can be avoided by administering anti-Rh antibodies to the mother immediately after the delivery of the first child.

C. Parasympathetic neural signals increase the rate of heart beat.

D. Heart is situated in the abdominal cavity in between the two lungs.

E. Coagulum is a dark reddish brown scum formed at the site of a cut or an injury over a period of time.

Choose the most appropriate answer from the options given below:

(1) A, B, and D only

(2) B, C, and E only

(3) C and D only

(4) A, B, C and E only